

A Convenient Synthesis of α -Substituted β,γ -Unsaturated Ketones and Esters via the Direct Addition of Substituted Allylic Zinc Reagents Prepared by Direct Insertion

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Dedicated to Professor Scott E. Denmark on the occasion of his 60th birthday

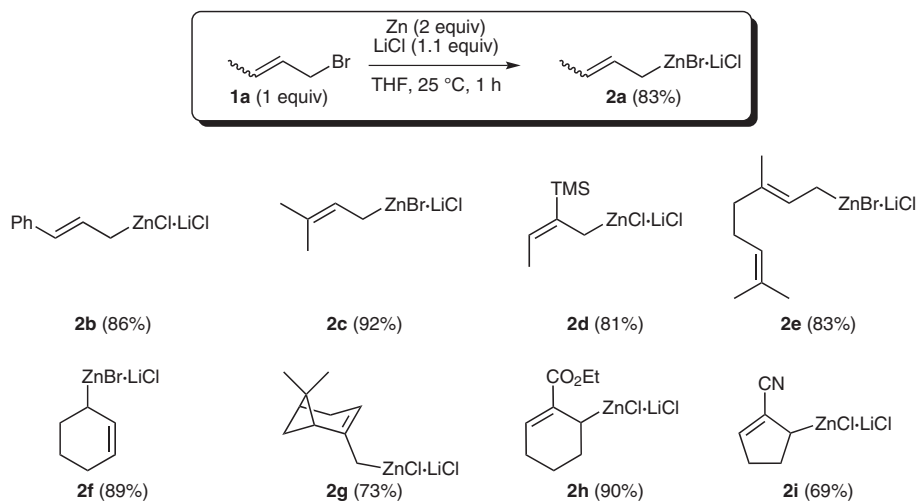
Abstract: A practical and convenient procedure for the synthesis of α -substituted β,γ -unsaturated ketones and esters has been developed. Substituted allylic zinc reagents, prepared via direct metal insertion in substituted allylic halides, react readily with a broad range of acid chlorides and chloroformates furnishing the corresponding α -substituted β,γ -unsaturated ketones and esters in high yield and perfect regioselectivity.

Key words: organozinc reagents, addition, unsaturated ketones and esters, insertion, allylic halides

The reaction of allylic organometallic reagents with carbonyl derivatives is of high importance in synthetic organic chemistry.¹ Especially allylic zinc reagents are very versatile organometallic species since their behavior is much more predictable than that of the corresponding allylic magnesium or lithium reagents.² Whereas allylzinc bromide in THF is readily available by the reaction of zinc foil with allyl bromide,³ the preparation of substituted zinc reagents is much more challenging due to competitive homocoupling reactions. Recently, we have reported the use of commercial zinc powder in the presence of lithium chloride in THF as a cheap and convenient method

for the synthesis of substituted allylic zinc reagents from allyl halides or phosphonates.⁴

β,γ -Unsaturated ketones are versatile building blocks in organic chemistry.⁵ Although a number of synthetic methods have been disclosed, only a few have been proven practical and useful. The acylation of olefins allows the synthesis of β,γ -unsaturated ketones, but generates α,β -unsaturated ketones as side-products.⁶ The reaction of various allylic organometallics with acyl halides is also reported in literature. Silicon,⁷ tin,⁸ copper,⁹ rhodium,¹⁰ manganese,¹¹ titanium,¹² mercury,¹³ cadmium,¹⁴ and indium¹⁵ are some of the metal powders reported earlier to synthesize β,γ -unsaturated ketones. But these protocols are mostly neither simple nor straightforward and are therefore of limited application. Since the reaction of allylic zinc reagents with acid chlorides¹⁶ or nitriles¹⁷ is a promising approach, we focused on the development of an addition reaction using the now readily available substituted allylic zinc reagents. Herein, we wish to report a simple and flexible synthesis of α -substituted β,γ -unsaturated ketones and esters through the addition of substituted allylic zinc reagents to a broad range of acid chlorides and chloroformates.



Scheme 1 LiCl-mediated preparation of substituted allylic zinc organometallics by direct insertion of zinc powder (yields determined by iodometric titration¹⁸)

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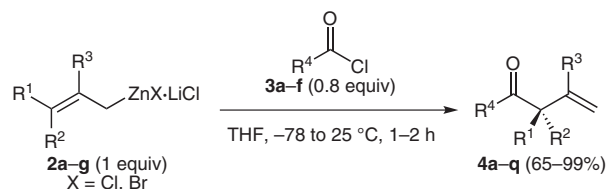
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Preliminary experiments had shown that the direct zinc insertion into allylic halides in the presence of lithium chloride provides the corresponding allylic zinc reagents almost without formation of homocoupling products. Thus, under optimized conditions but-2-en-1-ylzinc bromide (**2a**) was formed in one hour at 25 °C in 83% yield by dropwise addition of 1-bromobut-2-ene (**1a**; 1 equiv) to a suspension of commercial zinc powder (2.0 equiv) and dry lithium chloride (1.1 equiv) in THF (Scheme 1). This procedure was successfully extended to other allylic halides leading to cinnamylzinc chloride (**2b**; 86%), (3-methylbut-2-en-1-yl)zinc bromide (**2c**; 92%), (3,7-dimethylocta-2,6-dien-1-yl)zinc bromide (**2e**; 83%), and cyclohex-2-en-1-ylzinc bromide (**2f**; 89%). Especially the preparation of zinc reagent **2b** is remarkable, since cinnamyl chloride is known to readily undergo extensive homocoupling reaction during the synthesis of the corresponding zinc reagent. It is noteworthy that also functional groups like an ester or a nitrile were tolerated in the insertion reaction. Hence, 2-(ethoxycarbonyl)cyclohex-2-en-1-ylzinc chloride (**2h**) and 2-cyanocyclopent-2-en-1-ylzinc chloride (**2i**) were obtained from their corresponding chlorides in 90% and 69% yield, respectively. Starting from 2-chloromethyl-6,6-dimethylbicyclo[3.1.1]hept-2-ene (**1g**), the corresponding zinc reagent **2g** could be synthesized in 73% yield (25 °C, 30 h).^{4a} [2-(Trimethylsilyl)but-2-en-1-yl]zinc chloride (**2d**) was generated from its chloride in the presence of zinc powder (10 equiv) and lithium chloride (3 equiv) in 18 hours at 25 °C in 81% yield.^{4b}

We then decided to concentrate our studies to the addition of these highly reactive allylic zinc reagents to a broad

range of acid chlorides (Scheme 2 and Table 1). It turned out that this reaction proceeded under exceedingly mild conditions (−78 °C, 1 h) and furnished selectively β,γ -unsaturated ketones without any traces of the α,β -unsaturated isomers. Thus, the addition of but-2-en-1-ylzinc bromide (**2a**) to the (hetero)aromatic acid chlorides **3a–c** furnished selectively the corresponding α -substituted β,γ -unsaturated ketones **4a–c** in high yields (Table 1, entries 1–3). Regardless of the substitution pattern of the (hetero)aromatic acid chloride, the reaction proceeded in one hour at −78 °C. Noteworthy, the configuration of the double bond in the zinc reagent does not affect the reaction course, allowing the use of *E*- and *Z*-isomeric mixtures. Cinnamylzinc chloride (**2b**) reacted in a similar manner. The α -substituted β,γ -unsaturated ketones **4d–f** were obtained by addition to the corresponding (hetero)aromatic acid chlorides **3a**, **3b**, and **3d** in excellent yields (entries 4–6). Interestingly, also with the aliphatic acid chloride **3e** the addition proceeded smoothly (−20 °C → 25 °C, 2 h) and led to ketone **4g** in 72% yield (entry 7).



Scheme 2 Synthesis of α -substituted β,γ -unsaturated ketones of type **4** via addition of allylic zinc reagents **2** to various acid chlorides of type **3**

Table 1 Preparation of α -Substituted β,γ -Unsaturated Ketones of Type **4**

Entry	Substrate	Acid chloride	Product (Yield %) ^a	Conditions (Temp, Time)
1	 2a	 3a	 4a (85)	−78 °C, 1 h
2	2a	 3b	 4b (71)	−78 °C, 1 h
3	2a	 3c	 4c (65)	−78 °C, 1 h

Table 1 Preparation of α -Substituted β,γ -Unsaturated Ketones of Type 4 (continued)

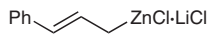
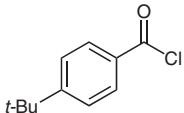
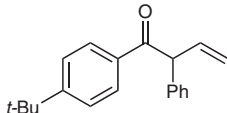
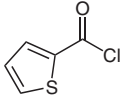
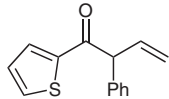
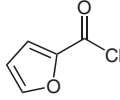
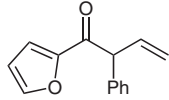
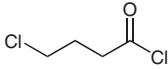
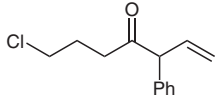
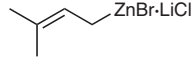
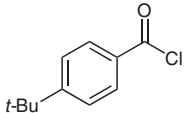
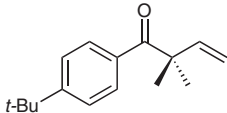
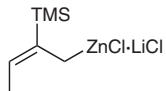
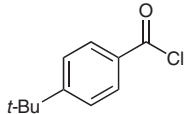
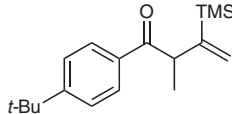
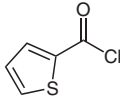
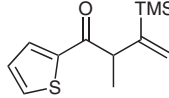
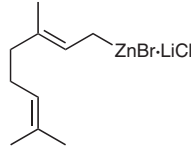
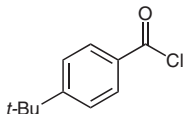
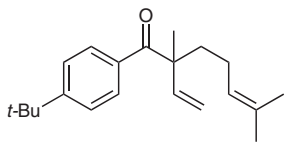
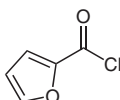
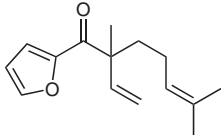
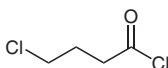
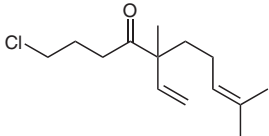
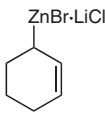
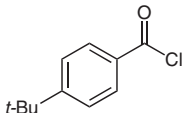
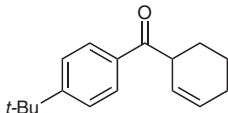
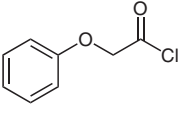
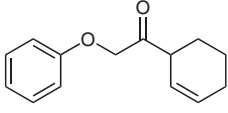
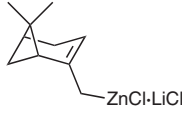
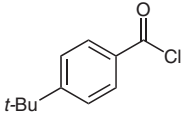
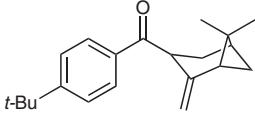
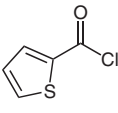
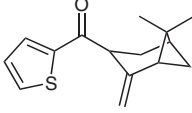
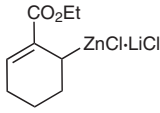
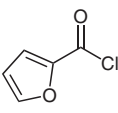
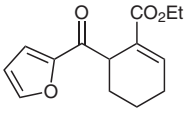
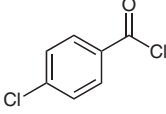
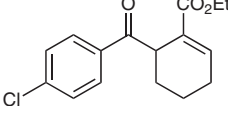
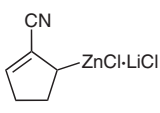
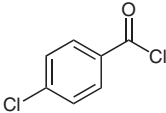
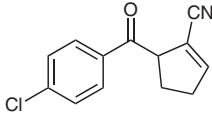
Entry	Substrate	Acid chloride	Product (Yield %) ^a	Conditions (Temp, Time)
4	 2b	 3a	 4d (90)	–78 °C, 1 h
5	2b	 3b	 4e (77)	–78 °C, 1 h
6	2b	 3d	 4f (84)	–78 °C, 1 h
7	2b	 3e	 4g (72)	–20 °C → 25 °C, 2 h
8	 2c	 3a	 4h (93)	–78 °C, 1 h
9	 2d	 3a	 4i (98)	25 °C, overnight
10	2d	 3b	 4j (99)	–78 °C, 1 h
11	 2e	 3a	 4k (70)	–78 °C, 1 h
12	2e	 3d	 4l (79)	–78 °C, 1 h
13	2e	 3e	 4m (74)	–78 °C, 1 h

Table 1 Preparation of α -Substituted β,γ -Unsaturated Ketones of Type **4** (continued)

Entry	Substrate	Acid chloride	Product (Yield %) ^a	Conditions (Temp, Time)
14			 4n (83)	–78 °C, 1 h
15	2f		 4o (75)	–78 °C, 1 h
16			 4p (67)	–78 °C, 1 h
17	2g		 4q (89)	–78 °C → 25 °C, 2 h
18			 4r (90)	–78 °C → 25 °C, overnight
19	2h		 4s (80)	–78 °C → 25 °C, overnight
20			 4t (70)	–78 °C → 25 °C, overnight

^a Yield of analytically pure isolated product as determined by ¹H NMR analysis.

This procedure could be successfully applied to trisubstituted allylic zinc derivatives. Thus, (3-methylbut-2-en-1-yl)zinc bromide (**2c**) reacted selectively in one hour at –78 °C with 4-(*tert*-butyl)benzoyl chloride (**3a**) to afford the corresponding α,α -disubstituted β,γ -unsaturated ketone **4h** in 93% yield (entry 8). Remarkably, the addition of [2-(trimethylsilyl)but-2-en-1-yl]zinc chloride (**2d**) to the acid chlorides **3a** and **3b** furnished the corresponding ketones **4i** and **4j** in almost quantitative yields of 98% and 99%, respectively (entries 9 and 10). Also the trisubstituted allylic zinc reagent **2e**, that contains another double bond besides the allylic one, could be employed in the addition reaction. (Hetero)aromatic acid chlorides **3a** and **3d** as well as an aliphatic one **3e** were used to synthesize the

corresponding ketones **4k–m** leaving the nonallylic double bond untouched (entries 11–13).

The cyclic allylic zinc reagents **2f–i** showed an analogous behavior. The addition reaction of cyclohex-2-en-1-ylzinc bromide (**2f**) and the acid chlorides **3a** and **3f** afforded selectively the corresponding α -substituted β,γ -unsaturated ketones **4n** and **4o** in high yields (83% and 75%, entries 14 and 15). Moreover, the cyclic zinc reagent **2g** reacted with the (hetero)aromatic acid chlorides **3a** and **3b** to the ketones **4p** and **4q** containing a terminal double bond in 67% and 89% yield, respectively (entries 16 and 17). Also, the zinc reagent **2h** underwent the addition reaction with the (hetero)aromatic acid chlorides **3d** and **3g** smoothly and furnished the corresponding ketones **4r** and **4s** in high yields (90% and 80%, entries 18 and 19). The

Table 2 Preparation of α -Substituted β,γ -Unsaturated Esters of Type 6

Entry	Substrate	Chloroformate	Product (Yield %) ^a	Conditions (Temp, Time)
1			 6a (64)	–78 °C, 1 h
2	2b		 6b (78)	–78 °C → 25 °C, 2 h
3	2b		 6c (82)	–78 °C → 25 °C, 2 h
4			 6d (70)	–20 °C → 25 °C, 2 h

^a Yield of analytically pure isolated product as determined by ¹H NMR analysis.

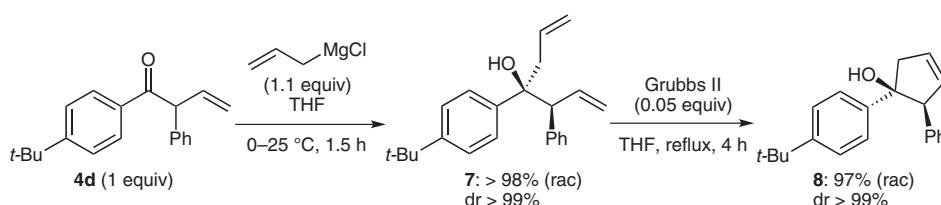
addition of 2-cyanocyclopent-2-en-1-ylzinc chloride (**2i**) to acid chloride **3g** led to the α -substituted β,γ -unsaturated ketone **4t** (70% yield, entry 20).

Due to the lack of a convenient and practical direct synthesis for α -substituted β,γ -unsaturated esters in the literature, our method was extended to this direction. As shown in Table 2, the allylic zinc reagents **2b** and **2c** reacted readily under optimized conditions with various chloroformates and formed the desired esters. Cinnamylzinc chloride (**2b**) added to aliphatic **5a**, aromatic **5b** as well as to allylic chloroformates **5c** affording the corresponding α -substituted β,γ -unsaturated esters in 64%, 78%, and 82% yield, respectively (Table 2, entries 1–3). The trisubstituted allylic zinc reagent **2c** showed a similar behavior and furnished ester **6d** in 70% yield (entry 4).

Ring-closing metathesis (RCM) represents one of the most powerful and versatile tool in organic synthesis for the formation of C=C bonds¹⁹ and has proven to be highly important for natural product synthesis.²⁰ With the α -sub-

stituted β,γ -unsaturated ketones and esters in hands, the diene precursor **7** for a RCM was readily synthesized in only one step via the diastereoselective addition¹ of allylmagnesium chloride to the carbonyl moiety of **4d** in almost quantitative yield (Scheme 3). The subsequent RCM using the second generation of Grubbs' catalyst²¹ furnished diastereoselectively the cyclopentene derivative **8** in 97% yield.

In summary, we have described a simple, convenient, and straightforward protocol for the selective synthesis of α -substituted β,γ -unsaturated ketones and esters via the direct addition of substituted allylic zinc reagents to a broad range of acid chlorides and chloroformates. Besides the convenient synthesis of the allylic zinc reagents through the lithium chloride-mediated zinc insertion in allylic halides, the notable advantages of this methodology are the mild reaction conditions, the short reaction times as well as the high yields. Further studies of this work are currently underway in our laboratory.

**Scheme 3** Diastereoselective addition of allylmagnesium chloride to **4d** and subsequent ring-closing metathesis forming the cyclopentene derivative **8**

For complete experimental procedures, analytical data, and NMR spectra see the Supporting Information.

All reactions were carried out under an argon atmosphere in flame-dried glassware. Syringes that were used to transfer anhydrous solvents or reagents were purged with argon prior to use. THF was continuously refluxed and freshly distilled from sodium benzophenone ketyl under N_2 . Yields refer to isolated yields of compounds estimated to be >95% pure as determined by 1H NMR (25 °C) analysis and capillary GC. Column chromatography was performed using silica gel (0.040–0.063 mm, 230–400 mesh ASTM) from Merck, unless otherwise indicated. All reagents were obtained from commercial sources unless stated otherwise.

Allylic Zinc Reagents via Direct Zinc Insertion in the Presence of Lithium Chloride; But-2-en-1-ylzinc Bromide (**2a**); Typical Procedure

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with Zn powder (1.31 g, 20 mmol) and LiCl (466 mg, 11 mmol). LiCl was dried in vacuo using a heat gun (450 °C, 5 min). After the addition of THF (10 mL), the Zn powder was activated with 1,2-dibromoethane (0.05 mL, 2 mol%) and Me_3SiCl (0.1 mL, 5 mol%). Subsequently, a solution of 1-bromobut-2-ene (**1a**; 1.35 g, 10.0 mmol) in THF (10 mL) was added dropwise at 25 °C and stirred for 1 h. Then, the remaining Zn powder was allowed to settle down and the supernatant solution was transferred into a dry, argon-flushed Schlenk flask. Iodometric titration¹⁸ furnished a concentration of 0.42 M (83% yield).

α -Substituted β,γ -Unsaturated Ketones via Direct Addition of Allylic Zinc Reagents to Acid Chlorides; 1-[4-(*tert*-Butyl)phenyl]-2-methylbut-3-en-1-one (**4a**); Typical Procedure

A dry, argon-flushed Schlenk flask equipped with a magnetic stirring bar and a septum was charged with a solution of 4-(*tert*-butyl)benzoyl chloride (**3a**; 315 mg, 1.6 mmol) in THF (1.5 mL) and cooled down to –78 °C. Subsequently, the freshly prepared allylic zinc reagent **2a** (4.75 mL, 2.00 mmol, 0.42 M in THF) was added dropwise and the reaction mixture was stirred at –78 °C for 1 h. The mixture was quenched with sat. aq. NH_4Cl (10 mL) and extracted with EtOAc (3 $\times$ 20 mL). The combined organic phases were dried (Na_2SO_4) and concentrated in vacuo. The crude residue obtained was purified by flash column chromatography (silica gel, pentane– Et_2O , 50:1) to give the analytically pure product **4a** as a colorless oil (294 mg, 85%).²²

IR (ATR): 3080, 3055, 3039, 2964, 2933, 2905, 2870, 1678, 1634, 1604, 1455, 1408, 1364, 1268, 1220, 1191, 1109, 993, 974, 963, 916, 847, 790, 719 cm^{-1} .

1H NMR (300 MHz, $CDCl_3$): δ = 7.96 (d, J = 8.6 Hz, 2 H), 7.50 (d, J = 8.9 Hz, 2 H), 5.95–6.11 (m, 1 H), 5.08–5.27 (m, 2 H), 4.10–4.25 (m, 1 H), 1.28–1.48 (m, 12 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 200.8, 156.7, 138.4, 133.7, 128.5, 125.5, 116.3, 45.4, 35.1, 31.1, 17.1.

MS (EI, 70 eV): m/z (%) = 216 (M^+ , 1), 162 (27), 161 (100), 160 (6), 146 (23), 118 (24), 115 (8), 91 (14).

HRMS: m/z calcd for $C_{15}H_{20}O$: 216.1514; found: 216.1501.

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Supporting Information for this article is available online at <http://www.thieme-connect.com/ejournals/toc/synthesis>.

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